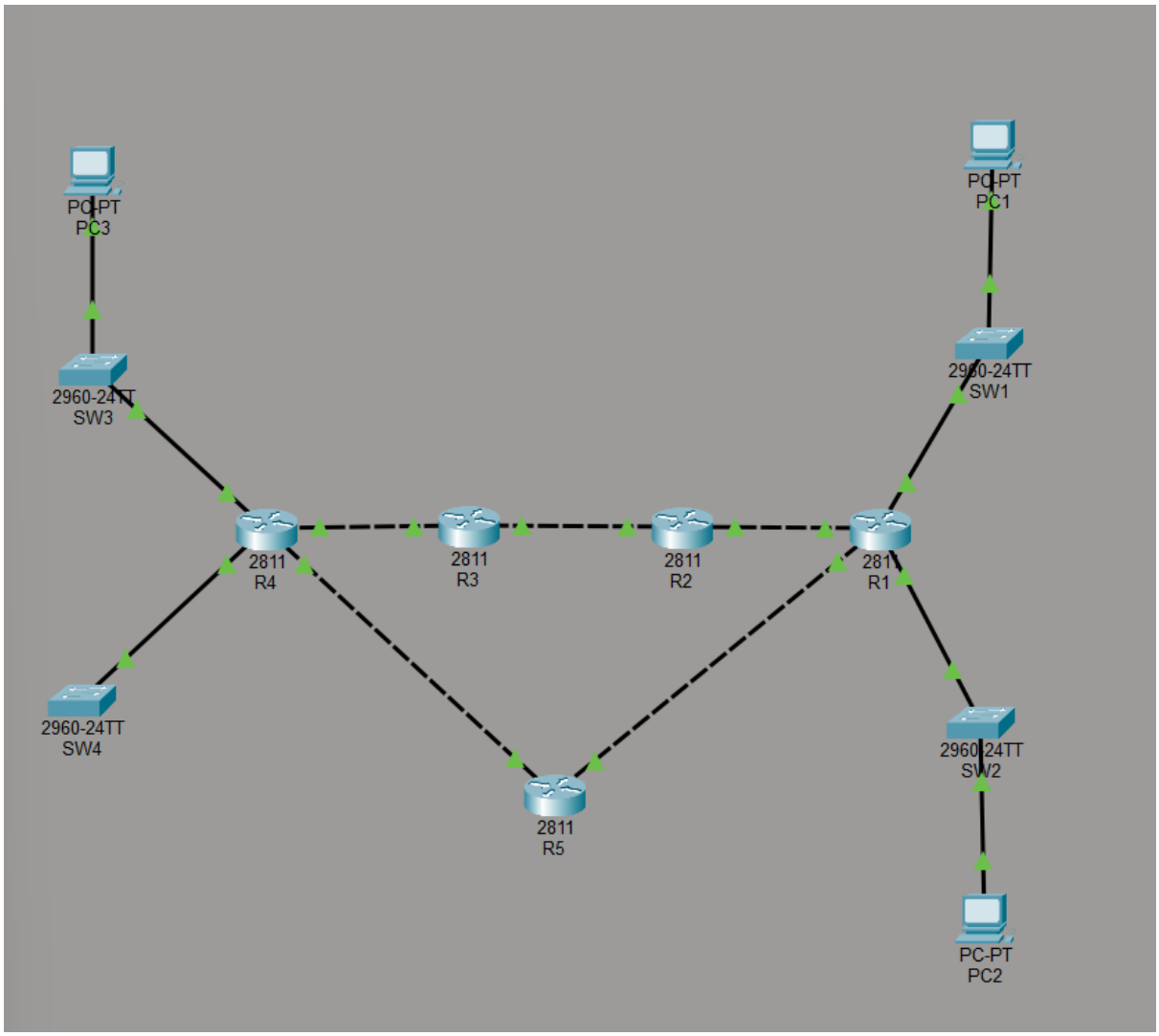
Network Topology



RIP Configuration on R1

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 10.0.0.0
R1(config-router)#end
R1#write memory
Building configuration...
[OK]
R1#show ip protocols
%SYS-5-CONFIG_I: Configured from console by console

Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 26 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0       22
  FastEthernet0/1       22
  FastEthernet1/0       22
  FastEthernet1/1       22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
          10.0.0.0
Passive Interface(s):
Routing Information Sources:
          Gateway         Distance      Last Update
Distance: (default is 120)
R1#
```

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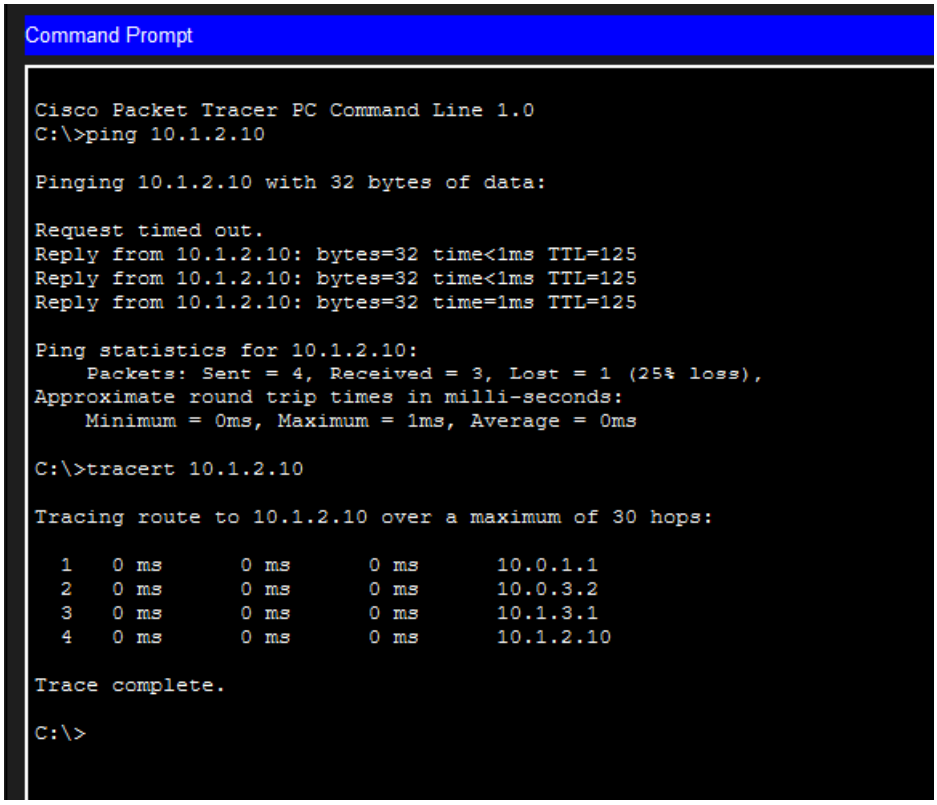
RIP Routing Table

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet1/0
L       10.0.1.1/32 is directly connected, FastEthernet1/0
C       10.0.2.0/24 is directly connected, FastEthernet1/1
L       10.0.2.1/32 is directly connected, FastEthernet1/1
C       10.0.3.0/24 is directly connected, FastEthernet0/1
L       10.0.3.1/32 is directly connected, FastEthernet0/1
S       10.1.0.0/24 [1/0] via 10.0.0.2
S       10.1.1.0/24 [1/0] via 10.0.0.2
R       10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:01, FastEthernet0/1
 --More--
```

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RIP Connectivity Test



RIPv1 Broadcast Updates

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 1
R1(config-router)#end
R1#debug ip rip
%SYS-5-CONFIG_I: Configured from console by console

RIP protocol debugging is on
R1#
```

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RIPv2 Multicast Updates

```
RIP protocol debugging is on
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#end
R1#debug ip rip
%SYS-5-CONFIG_I: Configured from console by console

RIP protocol debugging is on
R1#RIP: received v2 update from 10.0.0.2 on FastEthernet0/0
      10.1.0.0/24 via 0.0.0.0 in 1 hops
      10.1.1.0/24 via 0.0.0.0 in 2 hops
      10.1.2.0/24 via 0.0.0.0 in 3 hops
```

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RIP Learned Routes

```
R1#undebug all
All possible debugging has been turned off
R1#enable
R1#show ip route rip
     10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
R       10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:04, FastEthernet0/1
R       10.1.3.0/24 [120/1] via 10.0.3.2, 00:00:04, FastEthernet0/1

R1#
```

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RIP Database

```
R1#enable
R1#show ip rip database
10.0.0.0/24    auto-summary
10.0.0.0/24    directly connected, FastEthernet0/0
10.0.1.0/24    auto-summary
10.0.1.0/24    directly connected, FastEthernet1/0
10.0.2.0/24    auto-summary
10.0.2.0/24    directly connected, FastEthernet1/1
10.0.3.0/24    auto-summary
10.0.3.0/24    directly connected, FastEthernet0/1
10.1.0.0/24    auto-summary
10.1.0.0/24
    [1] via 10.0.0.2, 00:00:00, FastEthernet0/0
10.1.1.0/24    auto-summary
10.1.1.0/24
    [2] via 10.0.0.2, 00:00:00, FastEthernet0/0    [2] via 10.0.3.2, 00:00:02, FastEthernet0/1
10.1.2.0/24    auto-summary
10.1.2.0/24
    [2] via 10.0.3.2, 00:00:02, FastEthernet0/1
10.1.3.0/24    auto-summary
10.1.3.0/24
    [1] via 10.0.3.2, 00:00:02, FastEthernet0/1
R1#
```

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OSPF Routing Table

```
      P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet1/0
L       10.0.1.1/32 is directly connected, FastEthernet1/0
C       10.0.2.0/24 is directly connected, FastEthernet1/1
L       10.0.2.1/32 is directly connected, FastEthernet1/1
C       10.0.3.0/24 is directly connected, FastEthernet0/1
L       10.0.3.1/32 is directly connected, FastEthernet0/1
S       10.1.0.0/24 [1/0] via 10.0.0.2
S       10.1.1.0/24 [1/0] via 10.0.0.2
O       10.1.2.0/24 [110/4] via 10.0.0.2, 00:00:24, FastEthernet0/0
 --More--
```

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R1 successfully learned remote networks using OSPF, the letter O identifies OSPF routes, and the routing table displays the next hop addresses and path costs used to reach each network.

Verifying OSPF Routes

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet1/0
L       10.0.1.1/32 is directly connected, FastEthernet1/0
C       10.0.2.0/24 is directly connected, FastEthernet1/1
L       10.0.2.1/32 is directly connected, FastEthernet1/1
C       10.0.3.0/24 is directly connected, FastEthernet0/1
L       10.0.3.1/32 is directly connected, FastEthernet0/1
O       10.1.0.0/24 [110/2] via 10.0.0.2, 00:02:33, FastEthernet0/0
O       10.1.1.0/24 [110/3] via 10.0.0.2, 00:01:51, FastEthernet0/0
                    [110/3] via 10.0.3.2, 00:01:51, FastEthernet0/1
O       10.1.2.0/24 [110/3] via 10.0.3.2, 00:01:51, FastEthernet0/1
O       10.1.3.0/24 [110/2] via 10.0.3.2, 00:01:51, FastEthernet0/1

R1#
```

Copy Paste

R1 successfully learned remote networks through OSPF, the routing table shows two equal cost paths to the 10.1.1.0/24 network, demonstrating OSPF's ability to use multiple routes with the same cost.

Link Failure

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#interface fastethernet0/0
R2(config-if)#shutdown

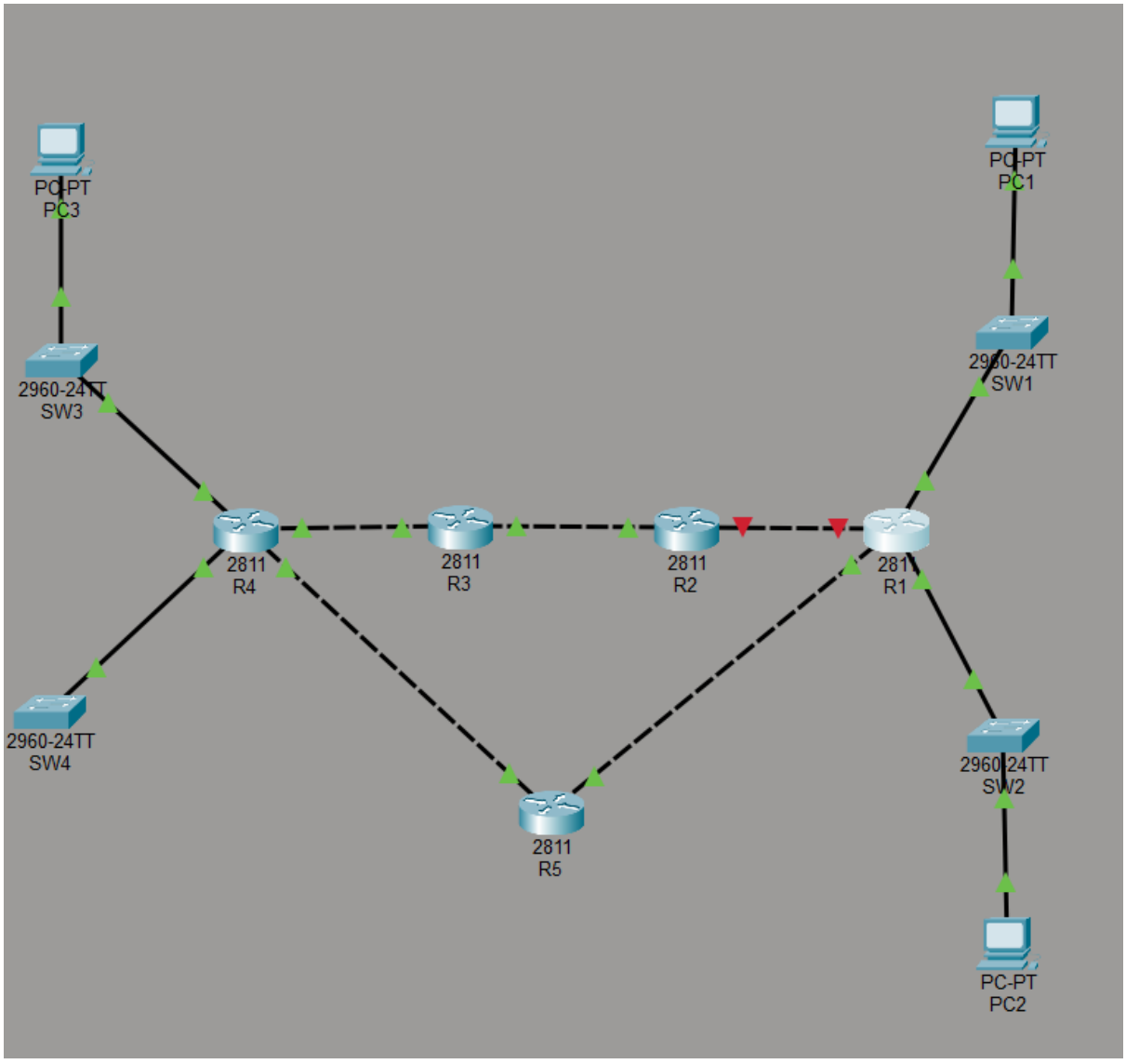
R2(config-if)#end
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down

00:54:39: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.3.1 on FastEthernet0/0 from FULL to DOWN, Neighbor Down:
Interface down or detached

R2#
%SYS-5-CONFIG_I: Configured from console by console
```

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OSPF After Link Failure

```
R1#enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
C       10.0.1.0/24 is directly connected, FastEthernet1/0
L       10.0.1.1/32 is directly connected, FastEthernet1/0
C       10.0.2.0/24 is directly connected, FastEthernet1/1
L       10.0.2.1/32 is directly connected, FastEthernet1/1
C       10.0.3.0/24 is directly connected, FastEthernet0/1
L       10.0.3.1/32 is directly connected, FastEthernet0/1
O       10.1.0.0/24 [110/4] via 10.0.3.2, 00:01:04, FastEthernet0/1
O       10.1.1.0/24 [110/3] via 10.0.3.2, 00:01:04, FastEthernet0/1
O       10.1.2.0/24 [110/3] via 10.0.3.2, 00:05:01, FastEthernet0/1
O       10.1.3.0/24 [110/2] via 10.0.3.2, 00:05:01, FastEthernet0/1

R1#
```

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After the connection between R1 and R2 was disabled, OSPF recalculated the network routes, R1 could still reach remote networks through R5, demonstrating OSPF's ability to adapt to network failures

OSPF Database

```
R1#enable
R1#show ip ospf database
            OSPF Router with ID (10.0.3.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
10.1.1.2        10.1.1.2        413         0x80000004 0x003cb6 2
10.1.3.2        10.1.3.2        383         0x80000004 0x00bc29 2
203.0.113.1     203.0.113.1     383         0x80000005 0x00a3ca 3
10.0.3.1        10.0.3.1        147         0x80000007 0x006479 3
10.1.0.2        10.1.0.2        147         0x80000005 0x00c15c 1

                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
10.1.0.1        10.1.1.2        433         0x80000001 0x007782
10.0.3.1        10.0.3.1        424         0x80000002 0x0057e6
10.1.1.1        203.0.113.1     413         0x80000001 0x00634d
10.1.3.1        203.0.113.1     383         0x80000002 0x00eb74
R1#
```

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Question 84

RIPv1 sends routing updates using broadcast to the address 255.255.255.255, while RIPv2 uses multicast to the address 224.0.0.9. RIPv2 also supports subnet mask information in its updates, making it more suitable for modern networks.

Question 85

RIP uses hop count to choose the best route to a destination. In our network, R1 has two paths to the 10.1.1.0/24 network: one through R2 and R3, and another through R5 and R4 so both paths have the same hop count, so RIP can use both routes for equal cost load balancing.

Question 86

RIP chooses the best route based on hop count, while OSPF uses cost, which is usually calculated from the bandwidth of the network links.

Question 87

After I shut down the FastEthernet0/0 interface on R2, the connection between R1 and R2 went down, OSPF automatically recalculated the routes and R1 started using R5 as its alternative path and the next hop address for the remote networks changed to 10.0.3.2.

Question 88

The RIP database shows the networks the router knows and how many hops it takes to reach them, the OSPF database has more information about the network and its connections, OSPF uses this information to find the best route.